Sample (i)	Feature 1 ($x_{i,1}$)	Feature 2 ($x_{i,2}$)	Actual Output (y_i)
1	1	2	5
2	2	1	4

Initial weights:

- **Weight 1 (w_1): 0.1**
- **Weight 2 (w_2): 0.2**
- **Bias (b): 0.0**
- **Equation: $\hat{y} = w_1 x_1 + w_2 x_2 + b$**

Sample (i)	$\hat{y}$	MSE	J
1	0.5	20.5	8.365
2	0.4	12.96	

$$w_1 \leftarrow w_1 - \alpha \frac{\partial L}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) x_{i,1}$$

$$w_2 \leftarrow w_2 - \alpha \frac{\partial L}{\partial w_2}$$

$$\frac{\partial L}{\partial w_2} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) x_{i,2}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{2} \left((0.5 - 5)(1) + (0.4 - 4)(2) \right)$$

$$= -5.85$$

$$w_1 = 0.1 - 0.1(-5.85)$$

$$= 0.685$$

$$\frac{\partial L}{\partial w_2} = \frac{1}{2}\left(0.5 - 5\right)(2) + \left(0.4 - 4\right)(1)$$

$$= -6.3$$

$$w_2 = 0.2 - 0.1(-6.3)$$

$$= 0.83$$

$$\frac{\partial L}{\partial b_1} = -4.05$$

$$b = 0 - 0.1(-4.05)$$

$$= 0.405$$

Parameter	Derivative Formula	Intuition
Weights (w_j)	$\frac{1}{n}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})x_j^{(i)}$	Error × Input Feature
Bias (b)	$\frac{1}{n}\sum_{i=1}^{n}(h_\theta(x^{(i)}) - y^{(i)})$	Just the Error